

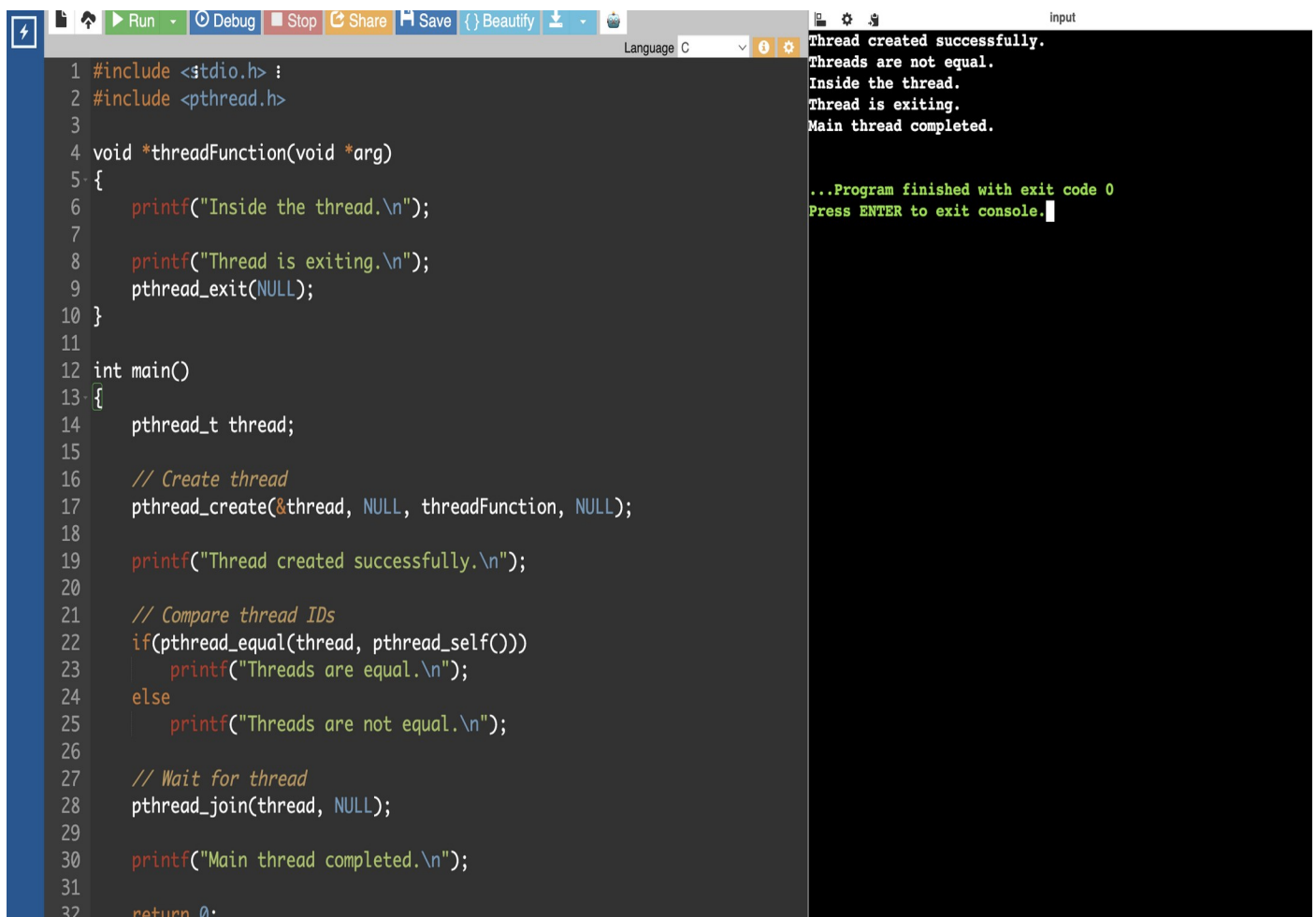
EXPERIMENT 30:

QUESTION:

Write C programs to demonstrate the following thread related concepts.

(i) create (ii) join (iii) equal (iv) exit

PROGRAM:



The image shows a code editor with a C program and its output. The code defines a thread function, creates a thread, compares thread IDs, joins the thread, and prints completion messages. The output shows the execution flow from thread creation to main thread completion.

```
1 #include <stdio.h> :
2 #include <pthread.h>
3
4 void *threadFunction(void *arg)
5 {
6     printf("Inside the thread.\n");
7
8     printf("Thread is exiting.\n");
9     pthread_exit(NULL);
10 }
11
12 int main()
13 {
14     pthread_t thread;
15
16     // Create thread
17     pthread_create(&thread, NULL, threadFunction, NULL);
18
19     printf("Thread created successfully.\n");
20
21     // Compare thread IDs
22     if(pthread_equal(thread, pthread_self()))
23         printf("Threads are equal.\n");
24     else
25         printf("Threads are not equal.\n");
26
27     // Wait for thread
28     pthread_join(thread, NULL);
29
30     printf("Main thread completed.\n");
31
32     return 0;
```

Thread created successfully.
Threads are not equal.
Inside the thread.
Thread is exiting.
Main thread completed.

...Program finished with exit code 0
Press ENTER to exit console.